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LLMs differ from human cognition because they are not embodied

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Large language models (LLMs) are impressive technological creations but they cannot replace all scientific theories of cognition. A science of cognition must focus on humans as embodied, social animals who are embedded in material, cultural and technological contexts.

In his famous (or infamous) 1972 book *What Computers Can't Do*[1], published early in the history of artificial intelligence (AI), Hubert Dreyfus argued that computers would never be intelligent because of the many ways that computers are unlike humans. More recently, writing in *The New Yorker*, Jaron Lanier claims that computers are just tools and that, as his title has it, "There is no artificial intelligence"[2]. These AI critics are insightful and hardly anti-technology. Dreyfus's book, in fact, has inspired many innovations in artificial intelligence, and Lanier has a long history as an innovator (including as a founding figure in virtual reality technology). However, despite their insights, Dreyfus and Lanier conflate two important yet different questions.

There is the technological question of whether computers can be intelligent, and also the scientific question of how it is that humans and other animals are intelligent. Answering either question requires an agreement about what the word 'intelligence' means. Here, I will both follow common usage and avoid making it a matter of definition that only adult humans could possibly be intelligent by assuming that to be intelligent is to have the ability to solve complex and cognitively demanding problems. If we understand intelligence this way, the question of whether computers can be intelligent has already been answered. With apologies to Dreyfus and Lanier, it has been clear for years that the answer is an emphatic 'yes'. The recent advances made by ChatGPT and other large language models (LLMs) are the cherry on top of decades of technological innovation. LLMs generate clear and convincing prose on almost any topic; they can be useful for research; they generate computer code that compiles and runs; and, recently, ChatGPT has been judged by patients to be as medically accurate as and more empathetic than human physicians[3]. I use ChatGPT regularly, as do many of my colleagues and students.

Of course, ChatGPT is far from perfect. For example, despite valiant efforts by their creators, it is easy to reveal the racial biases and sexism that infect the training sets of LLMs, which are scraped from the internet[4]. ChatGPT is also prone to doing what its creators spin as 'hallucinating', which is to say making things up whole cloth. Are hallucinations a sign that LLMs lack intelligence? I think not. Because LLMs generate responses to queries by producing statistically likely word sequences, they are not even trying to produce true statements about the world. LLMs are trying to solve the complex and cognitively demanding problem of generating clear sentences in response to queries, and in this task they are very successful. LLMs could be likened to politicians running for office. Both have a ready answer to any question; both also tend to make things up. Certainly, politicians are intelligent, and so are LLMs — even if both always require fact checking.

Yet although current LLMs may be intelligent, they are not intelligent in the same way as humans. For example, corpus studies show that children in the USA hear between 10 and 30 million spoken English words by their second birthdays (with a tragic correlation between socioeconomic status and the top and bottom ends of this range)[5]. But human children do not learn language the way that LLMs do. ChatGPT's Architecture 3.5 (the currently freely available version) was trained on 570 GB of text data, which is about 114 billion words. In other words, ChatGPT required a training set with more than 5,000 times the number of words that a typical child hears.

To answer the scientific question of how humans (and other animals) manage to be intelligent, we may need to start by asking how we differ from LLMs. Although humans are quite facile with and can learn quickly from text-based information (just as LLMs do), interacting with text is only one of our ways of knowing about the world around us. Reading and writing are evolutionarily recent, developmentally advanced human skills, and not skills that every human acquires. For nearly all the history of our species, humans — as with other animals — have known about the world exclusively through our embodied interaction with

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it and with our conspecifics; we mostly still know the world this way. This is the key to making sense of human intelligence, according to the contemporary embodied approach in the cognitive sciences[6], which denies that any computational system (including LLMs) could be the full answer to the scientific question. From this point of view, the key to a scientific theory of our intelligence lies in acknowledging the fact that humans are embodied, which is to say that we are living, biological creatures who are in constant interaction with the material, social, cultural and technological environment.

As living creatures, we have metabolic needs that are specific to the nature of our bodies. We have motor and perceptual systems tuned to help us to meet those metabolic needs. Frogs, for example, eat flies and they are masters at detecting and catching them, just as modern humans are good at finding grocery stores and restaurants. The ways in which living, biological creatures perceive and move are tightly intertwined; perceiving the world is for guiding action and often involves action. To experience what is going on around us, humans and other animals move: we move our eyes, we crane our necks and we walk over to get a better look at things. This activity is not ancillary to seeing what is around us; it is part of the seeing. In other words, from the embodied point of view, human vision is not the responsibility of the eyes or a brain region. The thing that sees the world includes a brain and eyes, of course, but those eyes are always moving (thanks to muscles) and are set in a head on a neck on the torso of an animal that is also moving. Explaining human intelligence is going to have to take this whole embodied system into account. This tight connection between experiencing and acting is a central feature of human intelligence and is something that LLMs lack entirely. The intelligence of LLMs is not embodied and, because of this, is not like ours.

Another point of difference is that although ChatGPT's operation and training may require an enormous (and undisclosed) amount of electricity, LLMs are not living things with metabolic needs as are animals. Our metabolic needs imply a built-in positive or negative valence to our experience of situations in the world[7]. Living things have needs and because of this some situations are experienced as being better than others. Even single-celled organisms respond differently to different concentrations of specific chemicals in their milieus, working to keep conditions within the constraints required for their viability. Human cognition is at its root a set of tools that we use to keep ourselves alive, and this is why we experience some situations as good and some as bad. We humans are motivated to be warm and fed and loved. This motivation pervades our experience and infects even the most apparently disinterested cognition.

The lives of humans are also necessarily social. We begin our lives inside another human and are born utterly helpless. We develop in a world alongside other humans, especially with immediate care-givers who provide us with the warmth, food and love required to keep us going. Part of the reason we can learn language more quickly than a LLM can is because for us words come along with facial expressions, intonations, gestures and the context of temporally extended interpersonal interactions. We also live in specific cultural and technological settings that shape and enable our activities. As embodied beings, we navigate complex interpersonal situations, and manage our facial expressions, intonations and gestures so that they fit in with the context and our role in it[8]. We enter the lecture hall differently when we are the lecturer than we do when we are audience members; we enter the lecture hall differently in Tokyo than we do in Toronto.

One of the key features of human intelligence is our ability to adapt our ways of thinking and ways of living to new technologies, with the written word as an early example. In 2023, smartphones and AI are ubiquitous. Smartphones and AI (including LLMs) change our cognition, for better and for worse, partly by changing the environment and partly by changing the cognitive skills that are required to interact with it successfully[9]. That cat is out of the bag and is not going back in.

Combined, these differences between LLMs and human cognition point to way of living and being intelligent that is founded on what John Haugeland called "giving a damn"[10]. Humans give a damn. Part of giving a damn is being committed to maintaining one's own existence and relationships with the world and others; another part is caring whether one is 'hallucinating' or telling the truth. ChatGPT is incapable of caring about these things because it is not an embodied creature, alive and participating in the world that it generates text about. ChatGPT responded to my queries about this issue by confirming that LLMs do not give a damn. To be sure, future AI models (maybe incorporating LLM technology) might be built so they do give a damn. What we learn from today's LLMs is that there is more than one way to be intelligent. Our way of being intelligent is not the only way and not their way. This takes nothing away from LLMs as a technological achievement but makes clear that they currently are not much help for answering scientific questions about the intelligence of humans and other animals. There is still plenty of work for cognitive scientists to do. Doing that work will require careful attention to our being embodied — living, moving, social and enculturated — creatures, creatures that give a damn.

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